Question 4 **(25 marks)**

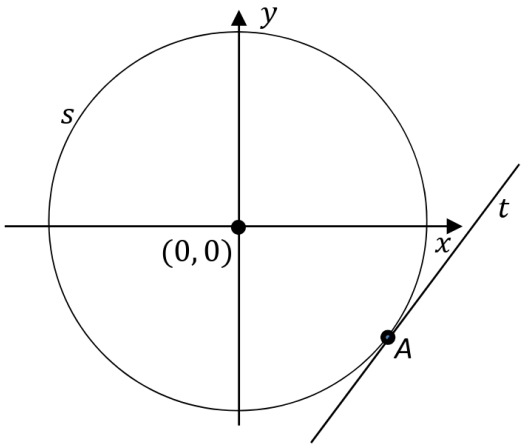
(a) **(i)** The circle c has equation $(x-1)^2+(y+4)^2=25$.
Find the centre and radius of c.

Centre: (,)

Radius: _____

(ii) The point $(1, k)$ is on c. Find the two possible values of k.

(b) The circle s has equation $x^2 + y^2 = 13$.
The point $A(3, -2)$ is on s.
Find the equation of t, the tangent to the circle at the point A.
Give your answer in the form
$ax + by + c = 0$, where $a, b, c \in \mathbb{Z}$.



Q4	Model Solution – 25 Marks	Marking Notes
(a) (i)	Centre $(1, -4)$ Radius 5	**Scale 10B (0, 5, 10)** *Partial Credit:* Work of merit • Centre or radius correct *Full Credit:* • Correct answer without supporting work
(a) (ii)	$(1-1)^2 + (k+4)^2 = 25$ $(k+4)^2 = 25$ $k^2 + 8k - 9 = 0$ $(k-1)(k+9) = 0$ $k = 1 \quad k = -9$ Or $(k+4)^2 = 25$ $k + 4 = \pm 5$ $k = 1 \quad k = -9$	**Scale 5D (0, 2, 3, 4, 5)** *Low Partial Credit:* Work of merit • Formula with some substitution *Mid Partial Credit:* • Formula fully substituted $(1-1)^2 + (k+4)^2 = 25$ *High Partial Credit:* Significant work • $k^2 + 8k - 9 = 0$ • $k + 4 = \pm 5$ • Substituting $(k, 1)$ into formula and completed correctly

(b)	$m_{Radius} = -\frac{2}{3}$ $\Rightarrow$ Perpendicular slope $= \frac{3}{2}$ $y + 2 = \frac{3}{2}(x - 3)$ $2y + 4 = 3x - 9$ $3x - 2y - 13 = 0$ or $(x-h)(x_1-h) + (y-k)(y_1-k) = r^2$ $(x-0)(3-0) + (y-0)(-2-0) = 13$ $3x - 2y - 13 = 0$	**Scale 10D (0, 3, 5, 8, 10)** *Low Partial Credit:* Work of merit • Formula with some substitution *Mid Partial Credit:* • m_R and/or m_T found • HL Formula with any <u>2</u> of the <u>3</u> elements substituted correctly *High Partial Credit:* • Eq. of line formula fully substituted • One incorrect substitution followed by correct solution **Note**: Accept $(-3x + 2y + 13 = 0)$

Question 7 **(55 marks)**

Cones is a sculpture in the National Gallery of Australia. It consists of 14 identical steel cones arranged into pairs which are joined together as shown.

Picture: http://cs.nga.gov.au

(a) **(i)** The height of each cone is equal to the diameter of its base. If the radius of the base is 2·25 m, write the height of a cone.

(ii) Show that, correct to 2 decimal places, the slant height, l, of a cone is 5·03 m.

(b) In order to maintain the steel's reflective shine, the surface is polished regularly.

(i) Find the curved surface area of the entire sculpture (14 cones).
Give your answer correct to 2 decimal places.

(ii) One litre of polish will cover 12·25 m^2. Find how many litres are needed to polish the entire sculpture. Give your answer correct to the nearest litre.

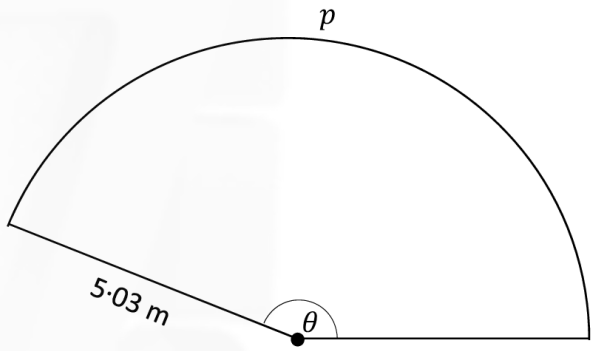
(iii) A container of polish contains 5 litres and costs A$110. Find the number of containers of polish that must be purchased in order to polish the entire sculpture and hence find the cost of the polish in euro (A$1 = €0·68). Give your answer correct to the nearest euro.

No. of containers:

Cost (€):

(c) The diagram (not to scale) shows the net of the outer surface of one of the cones of the sculpture. It is a sector of a circle of radius length 5·03 m with arc length p m and angle θ at the centre, as shown below.

(i) Find p, the length of the arc of the sector. Give your answer correct to 2 decimal places.



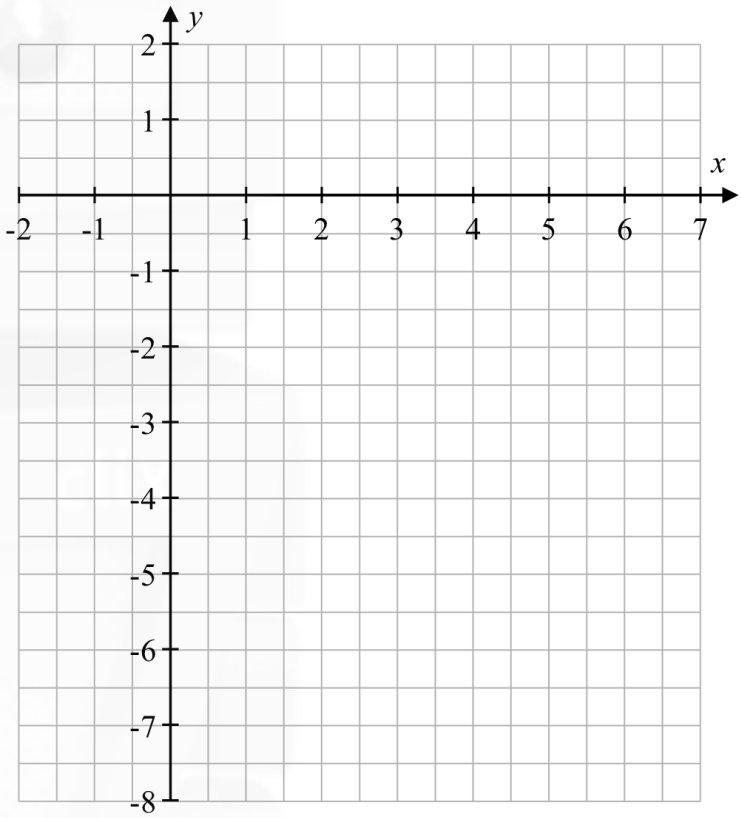
(ii) Find θ, the angle at the centre of the sector. Show all your working out. Give your answer correct to the nearest degree.

(a) **(i)**	$h = 2r$ $= 2(2{\cdot}25)$ $= 4{\cdot}5$ m	**Scale 10B (0, 5, 10)** *Partial Credit:* • Correct relationship identified. • $2{\cdot}25 \div 2$ • $h = d$ drawn on a diagram. *Full Credit:* • Correct answer without work.
(a) **(ii)**	$l^2 = h^2 + r^2$ $l^2 = (4{\cdot}5)^2 + (2{\cdot}25)^2$ $l^2 = 25{\cdot}3125$ $l = \sqrt{25{\cdot}3125}$ $= 5{\cdot}03$	**Scale 5D (0, 2, 3, 4, 5)** *Low Partial Credit:* • Some use of Pythagoras. • Work of merit. [e.g. Work with 2·25 / 4·5]. *Mid partial credit:* • Uses (5·03)to give: $(5{\cdot}03)^2 = (4{\cdot}5)^2 + (2{\cdot}25)^2$ *High Partial Credit:* • Pythagoras fully substituted. • Finishes with only one error. • Stops at $\sqrt{25{\cdot}3125}$
(b) **(i)**	CSA$= 14 \times \pi rl$ $= 14 \times \pi\,(2{\cdot}25)(5{\cdot}03)$ $= 497{\cdot}77$ m^2	**Scale 10C (0, 3, 7, 10)** *Low Partial Credit:* • CSA formula written with some correct substitution. • Work of merit. [e.g. Use of 2·25/5·03]. • Indicates multiplication by 14. *High Partial Credit:* • Fully correct substitution. • Finishes with only one error. • CSA of one cone calculated (35·55).

(b) (ii)	No. of litres $= \frac{497{\cdot}77}{12{\cdot}25} = 40{\cdot}63428571$ $= 41$	**Scale 10B (0, 5, 10)** *Partial Credit:* Works with answer to (b) (i). Works with $12{\cdot}25$.
(b) (iii)	Number of containers = $\frac{41}{5} = 8{\cdot}2 = 9$ $\text{Cost} = 9 \times 110$ $= A\$\,990$ $= 990 \times €0{\cdot}68$ $= €673{\cdot}20$ $= €673$	**Scale 10D (0, 4, 5, 7, 10)** *Low Partial Credit:* • Indicates answer as $\frac{(b)(ii)}{5}$ and stops . *Mid Partial Credit:* • Finds correct number of containers. (without work) *High Partial Credit:* Finds $A\$990$ or consistent equivalent. Finishes with only one error.
(c) (i) + (c) (ii)	$Arc\ length = 2\pi r$ $= 2\pi\,(2{\cdot}25)$ $= 14{\cdot}14\ \text{m}$ $Arc\ length = \frac{\theta}{360} \times 2\pi r$ $\frac{\theta}{360} \times 2\pi\,(5{\cdot}03) = 14{\cdot}14$ $\theta = \frac{14{\cdot}14 \times 360}{2\pi\,(5{\cdot}03)}$ $\theta = 161°$	**Scale 10C (0, 3, 7, 10)** *Low Partial Credit:* • Correct relevant formula written with some correct substitution. • Identifies r = $2{\cdot}25$ or CSA= $(35{\cdot}55)$. • Correct use of $360°$ *High Partial Credit:* • One correct. • Uses $\boldsymbol{l = r\theta}$ as degrees) ($\theta = 2{\cdot}8$). (1 rad $\approx 57{\cdot}296°$) • Substantial work in both parts.

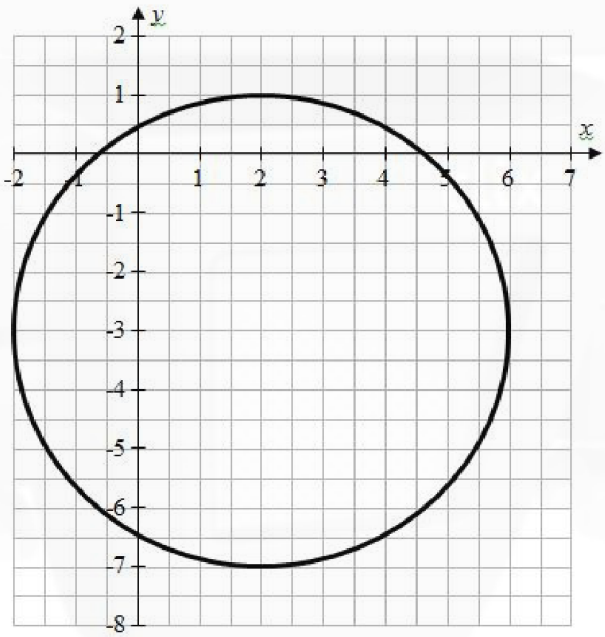
(a) The circle c has centre $(2, -3)$ and a radius of 4 cm.
Write down the equation of c.

(b) Draw the circle c on the grid opposite. Each unit on the co-ordinate grid is 1 cm.



(c) Verify, using algebra, that the point $(3, 1)$ is outside of c.

(d) Find the area of the smallest four-sided figure that will fit around the circle c.

Q3	Model Solution – 25 Marks	Marking Notes
(a)	$(x-2)^2+(y+3)^2=4^2$	Scale 5C(0, 1, 2, 5) *Low Partial Credit:* • Identifies correct formula. • Labels any one of (2, –3) or 4 correctly. • Works to $r^2=4^2$. *High Partial Credit:* • Answer as $(x-2)^2+(y-3)^2=4^2$. • Substitutes centre or radius correctly into formula.
(b)		Scale 5C(0, 1, 2, 5) *Low Partial Credit:* • Any work of merit e.g. Marks centre on grid. *High Partial Credit:* • Any TWO of the three components, centre, radius or drawing correct.
(c)	$(x-2)^2+(y+3)^2=4^2$ $(3-2)^2+(1+3)^2$ $1+16>16$ or $\sqrt{(3-2)^2+(1+3)^2}$ $\sqrt{17}>4$	Scale 10C(0, 2, 4, 10) *Low Partial Credit:* • Plots (3,1). • Identifies correct relevant formula. • Substitutes (3, 1) into correct/consistent formula . *High Partial Credit:* • $(1+16)$ or 17 but no conclusion. (>16) • $\sqrt{17}$ but no conclusion. (>4)
(d)	$8\times 8=64\text{ cm}^2$	Scale 5C(0, 1 ,2, 5) *Low Partial Credit:* • Drawing on grid. (Any one dimension) *High Partial Credit:* • Identifies correct/consistent square. • Scale error in dimensions calculated correctly.